Dohyun (Eric) Kim

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PROFILE

Penultimate Computer Systems Engineering (Hons) student at the University of Auckland, specialising in embedded systems and digital hardware design. AWS certified in both Cloud and AI/ML, with a competition-winning project portfolio spanning FPGA development, PCB design, and embedded robotics systems.

EDUCATION

University of Auckland – Auckland CBD, New Zealand — *Expected Graduation Nov 2027*

- Degree: *Bachelor of Engineering (Honours) in Computer Systems Engineering*
- Concentration: Embedded Systems & Digital Hardware Design

EXPERIENCES

Teaching Assistant — **Auckland, New Zealand**
University of Auckland — *Jun 2026 - Present*

- Delivered teaching support across tutorials, lab sessions, and test invigilation for **ELECTENG 292: Electronics** to strengthen **student leadership** and technical laboratory techniques.

Robotics Instructor — **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) — *Apr 2026 - Present*

- Instructed students in robot **hardware assembly** and **software programming** to complete missions, while coaching and supporting teams in preparation for **nationwide robotics competitions**.

PROJECTS

Flappy Universe — **VHDL**
FPGA Game Implementation - Team Project — *Jun 2026*

- Implemented a Flappy Bird-style game in VHDL on an **Altera FPGA** with VGA output and sprite rendering in order to demonstrate real-time hardware design.
- Developed a pixel-priority VHDL pipeline in order to render game scenes at VGA resolution.

Smart Energy Monitor — **Embedded C**
Embedded Energy Monitoring System - Team Project — *Oct 2025*

- Designed an embedded energy monitoring system using **ATmega328P** microcontrollers with ADC processing in order to measure and display real-time energy usage.
- Produced a validated PCB schematic using **Altium Designer** in order to achieve accurate signal conditioning.

AWARD

3rd Place in 2025 ECSE Design Competition — **Team Project**
“Winnie the Bot” - Interactive companion robot — *Sep 2025*

- Built an AI-powered robot using dual **ATmega328P** microcontrollers, an AI camera, and audio peripherals in order to enable face tracking, arm movement, and voice dialogue in a compact form factor.
- Contributed to **prototyping** the enclosure using iterative design in order to deliver a functional physical build.

CERTIFICATIONS

AWS Certified Cloud Practitioner — **Cloud Computing**
Amazon Web Services (AWS) — *Apr 2026*

- Validated foundational knowledge of **AWS cloud concepts**, core services, and cloud security and architecture.

AWS Certified AI Practitioner — **AI/ML**
Amazon Web Services (AWS) — *May 2026*

- Validated foundational knowledge of AI/ML concepts, generative AI, and **AWS AI/ML services** and tools.

LEADERSHIPS

Academic Team Executive — **Auckland, New Zealand**
KEB (Korean Engineering Body) — *Jul 2024 - Present*

- Tutored junior engineering students and assisted in planning academic events for the student community.

Full time Volunteer — **Auckland, New Zealand**
IEEE (Institute of Electrical and Electronics Engineers) — *Jul 2025*

- Volunteered 40+ hours full-time supporting competition operations at the NZ Robotics Olympiad 2025.

SKILLS

Languages & Frameworks: C, VHDL, Python, MATLAB, Java, R, React.js
Tools & Platforms: Altium Designer, LTSpice, ModelSim, Intel Quartus Prime, Proteus, Atmel AVR, Git, AutoCAD, AWS